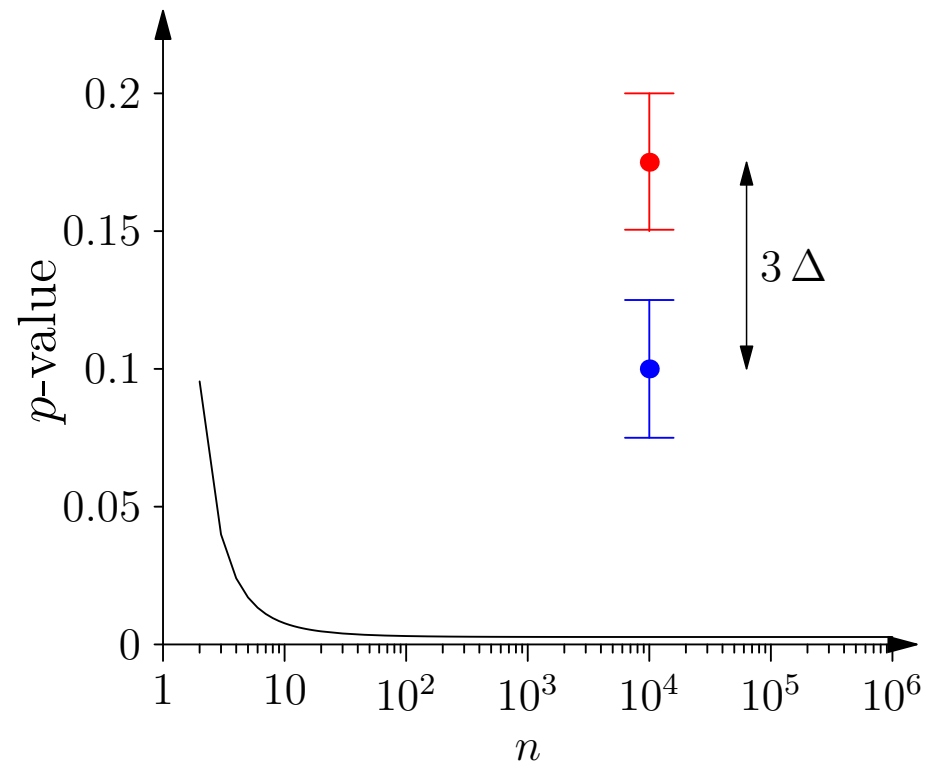
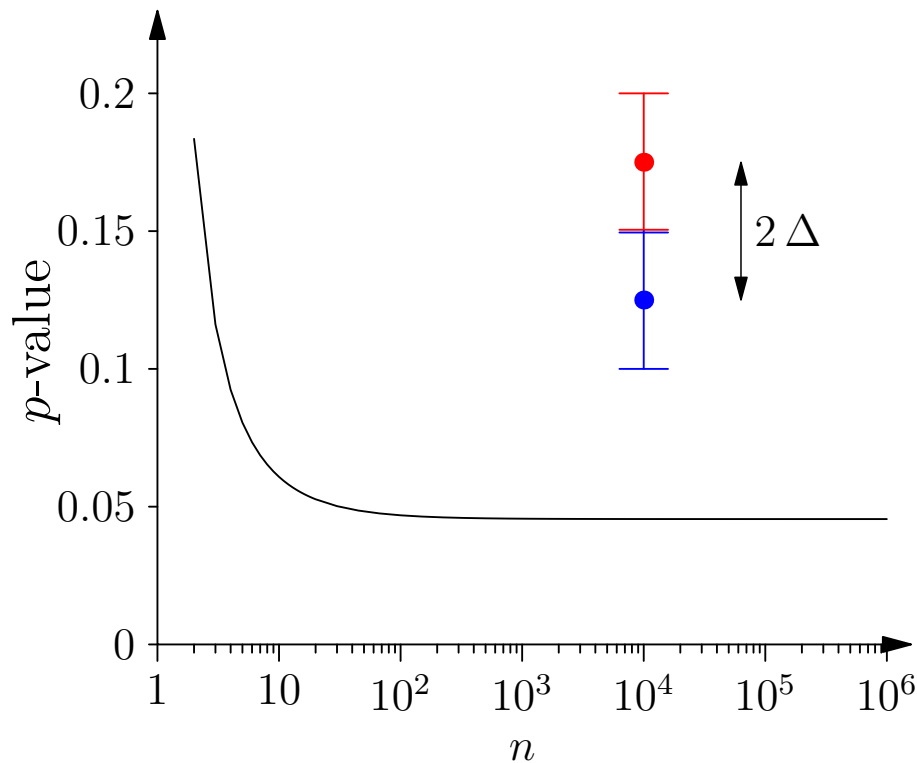


Mathematics of Machine Learning

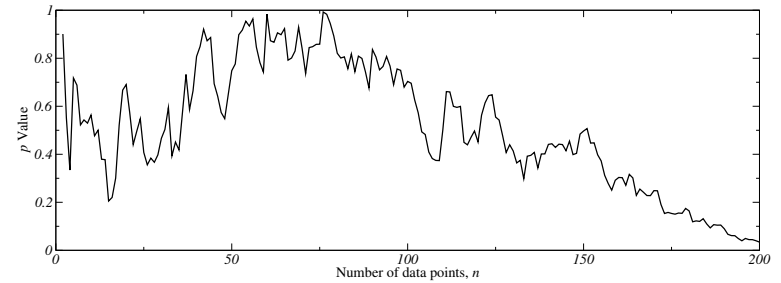
On Becoming a Scientist



interpreting data, standard errors, confidence tests

Outline

1. **Outline**
2. Experimental Error
3. Confidence Estimates



Experiments and Reproducibility

- One of the fundamental requirements of science is that experiments can be reproduced
- This applies to machine learning and deep-learning as it does to particle physics, molecular biology, etc.
- We need to describe completely what procedures were carried out
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Describing Your Experiments

- You need to tell the reader
 - ★ What your deep-learning architecture looks like
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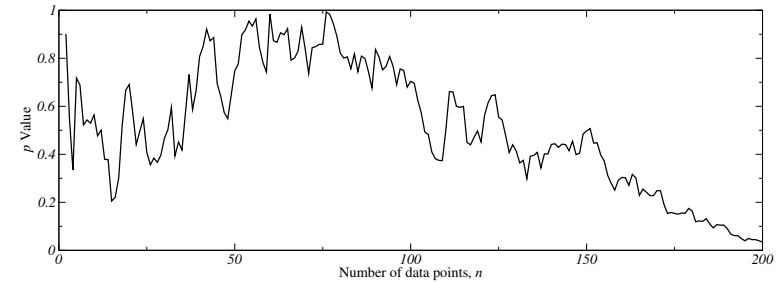
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- We assume that our test set consists of typical examples that we might encounter and that they are independent (e.g. we are not seeing the same example over and over again)
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Collecting Statistics

- Given some experimental measurements $\{X_i | i = 1, 2, \dots, n\}$ we can estimate the mean and variance

$$\hat{\mu}_n = \frac{1}{n} \sum_{i=1}^n X_i \qquad \hat{\sigma}_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \hat{\mu}_n)^2$$

- These are not the true mean, $\mu = \mathbb{E}[X_i]$, and true variance $\sigma^2 = \mathbb{E}[(X_i - \mu)^2]$, but they are unbiased estimates in that

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Repeated Bernoulli Trials

- Suppose we are measuring a binary outcome, success or failure
- Assuming the trials are independent (so called Bernoulli trials) then the only unknown is the probability, p , of a success
- If we make n trials then probability of k successes is given by the binomial distribution

$$\mathbb{P}(K = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

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- Assuming the trials are independent (so called Bernoulli trials) then the only unknown is the probability, p , of a success
- If we make n trials then probability of k successes is given by the binomial distribution

$$\mathbb{P}(K = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

- The mean and variance of K are

$$\mathbb{E}[K] = np \qquad \mathbb{E}[(K - np)^2] = np(1 - p)$$

Mean Squared Error

- For the case of n Bernoulli trials we can estimate the mean success rate

$$\hat{p} = \frac{K}{n}$$

- The mean squared error (MSE) in this estimate is

$$\begin{aligned}\mathbb{E}[(\hat{p} - p)^2] &= \mathbb{E}\left[\left(\frac{K}{n} - p\right)^2\right] = \frac{1}{n^2}\mathbb{E}[(K - np)^2] \\ &= \frac{1}{n^2}np(1 - p) = \frac{p(1 - p)}{n}\end{aligned}$$

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Standard Error for Bernoulli Trials

- The estimated error in the mean is given by the root mean squared error (RMSE) $\sqrt{\mathbb{E}[(\hat{p} - p)^2]} = \sqrt{\frac{p(1-p)}{n}}$
- We don't know p , but we can use an estimate, $\hat{p} = \frac{K}{n}$

$$\Delta \approx \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

- We can also use a worst-case bound (when $p = \frac{1}{2}$)

$$\Delta \approx \frac{1}{2\sqrt{n}}$$

This is pessimistic if p is small, although the previous estimate is optimistic if $\hat{p} = 0$

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- Traditionally errorbars show the estimated error in the mean (standard error, $\Delta = \hat{\sigma}_n / \sqrt{n}$)
- This means that about 30% of the time the true mean would be outside the errorbar
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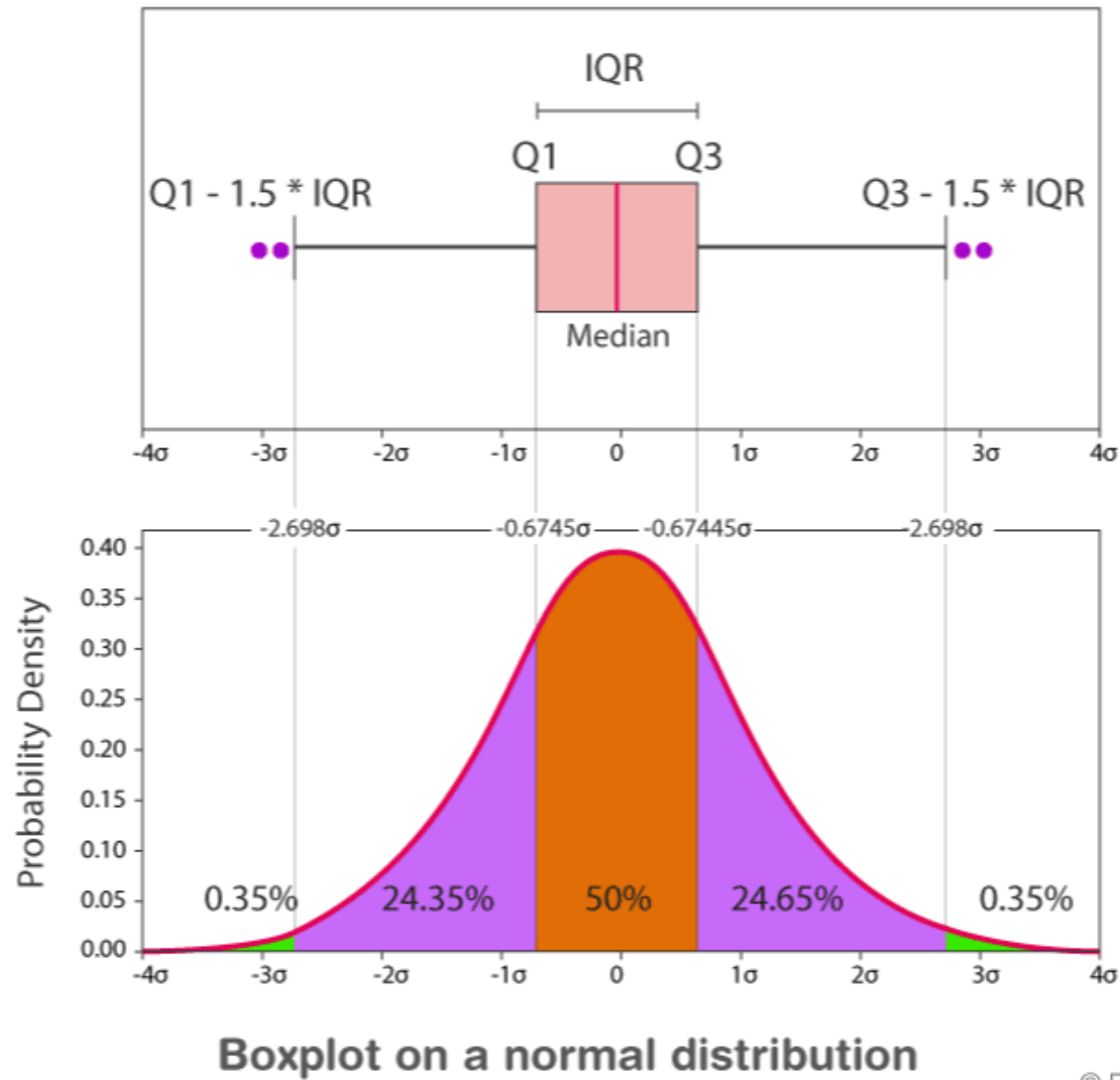
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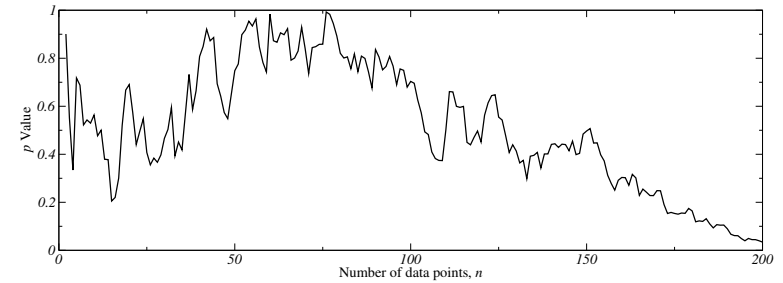
Box Plots



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Outline

1. Outline
2. Experimental Error
3. **Confidence Estimates**

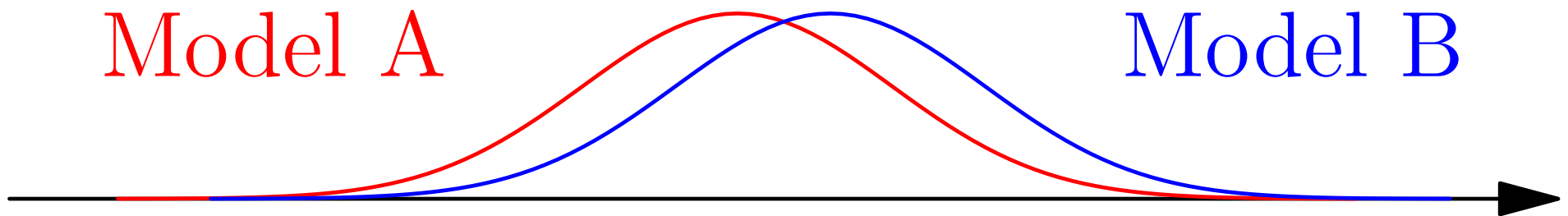


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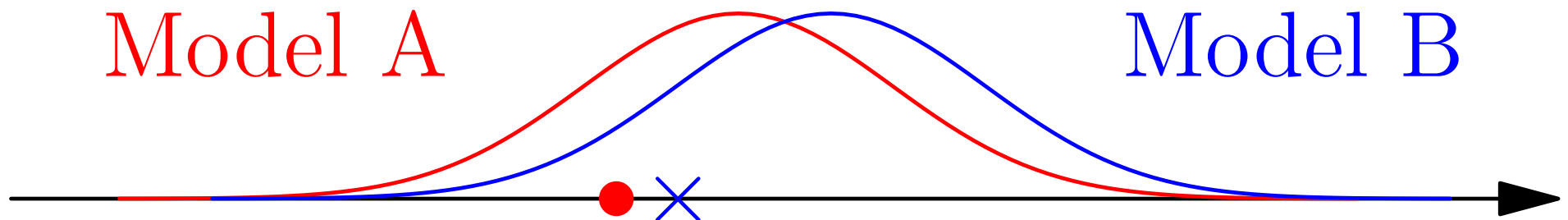
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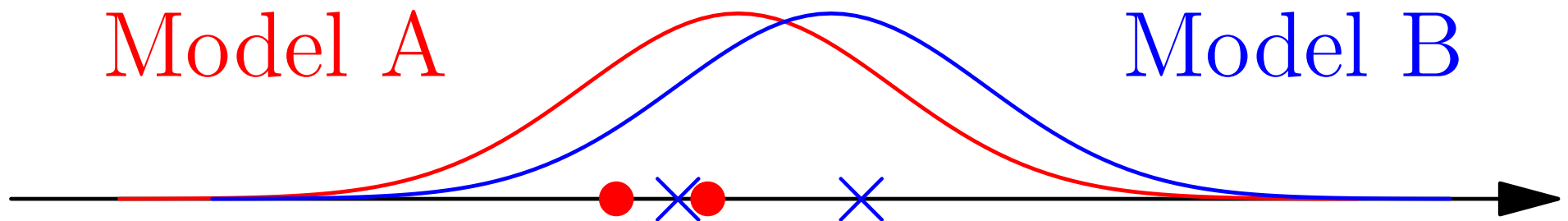
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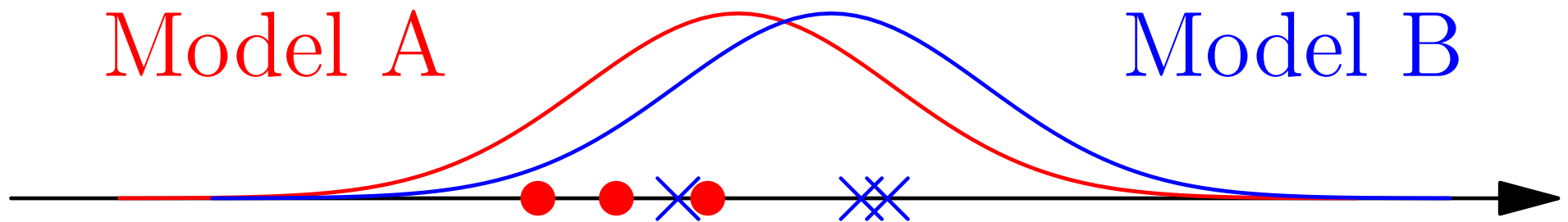
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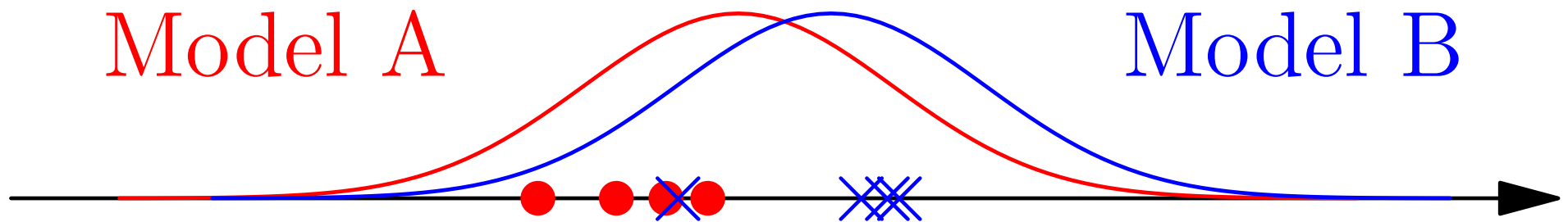
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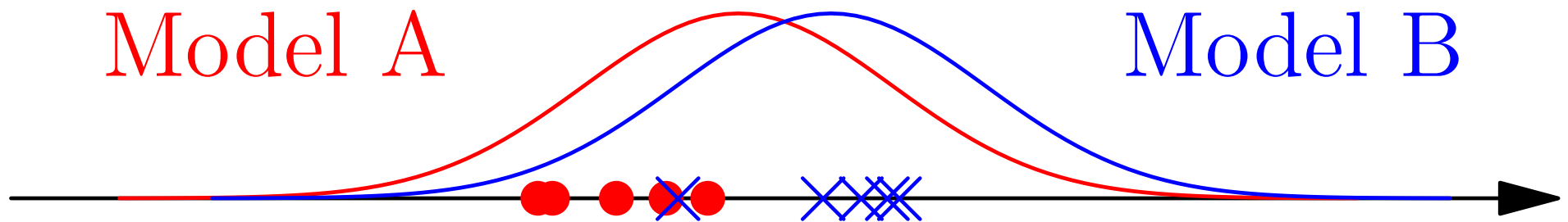
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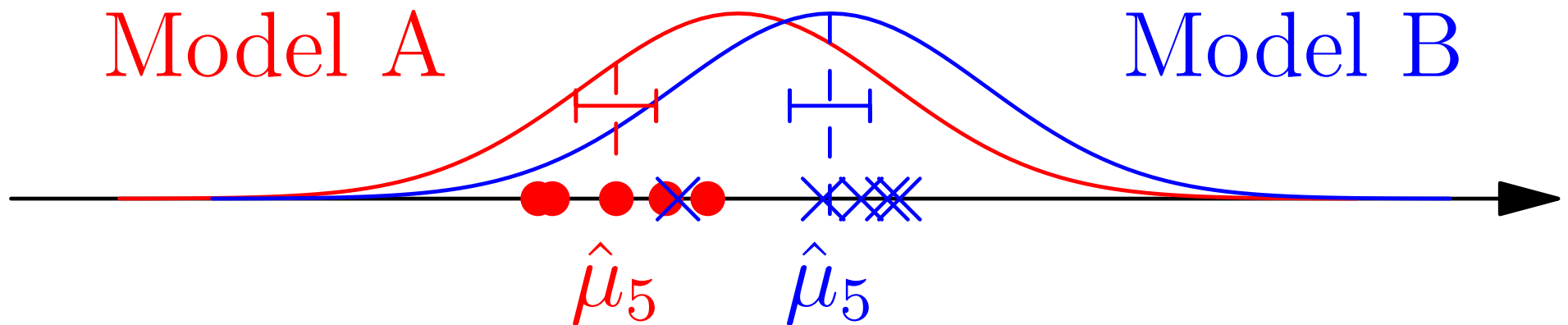
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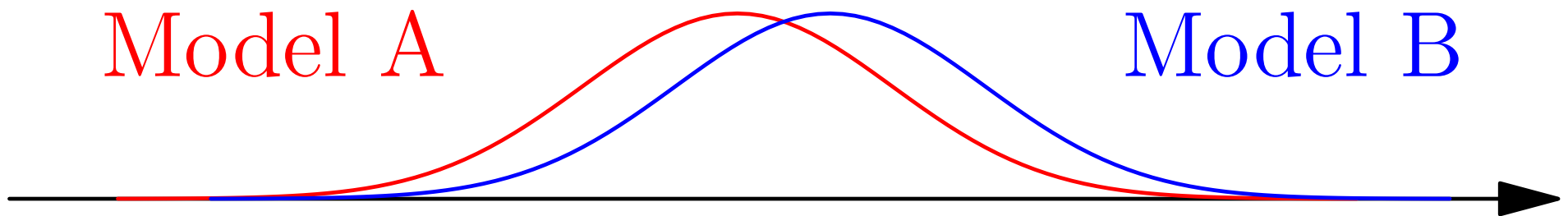
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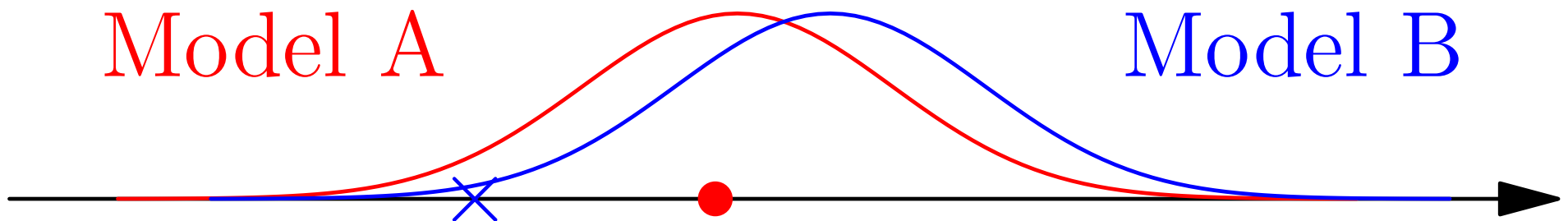
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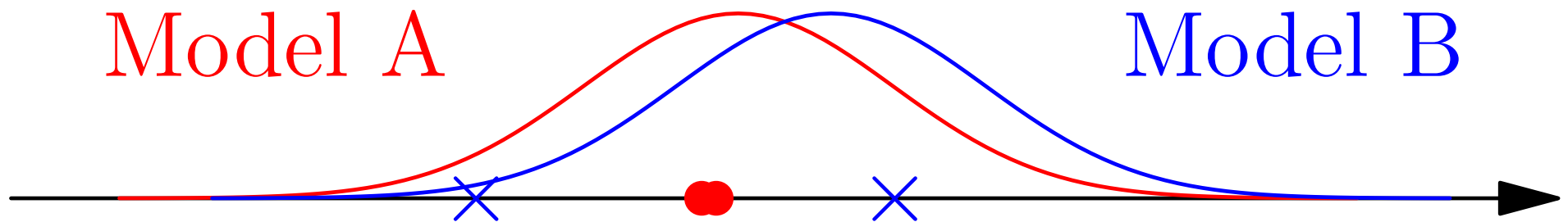
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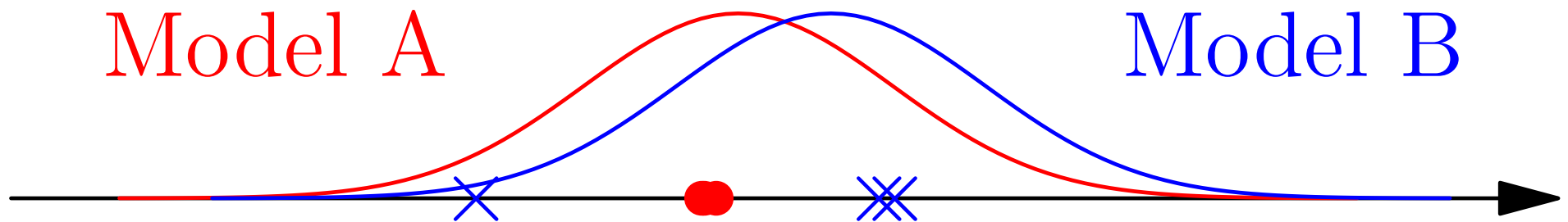
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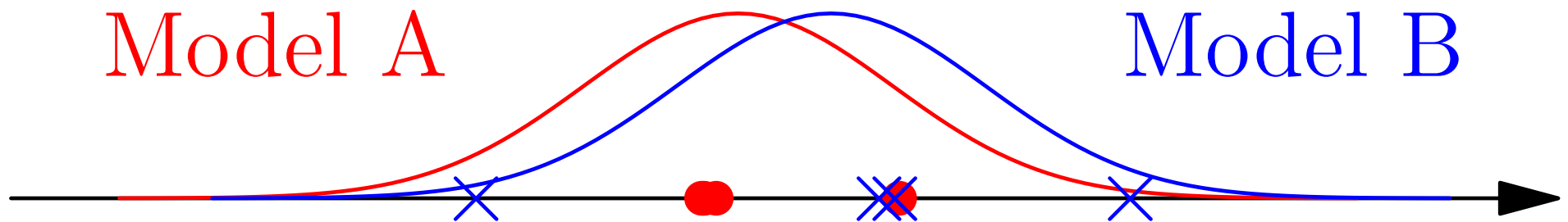
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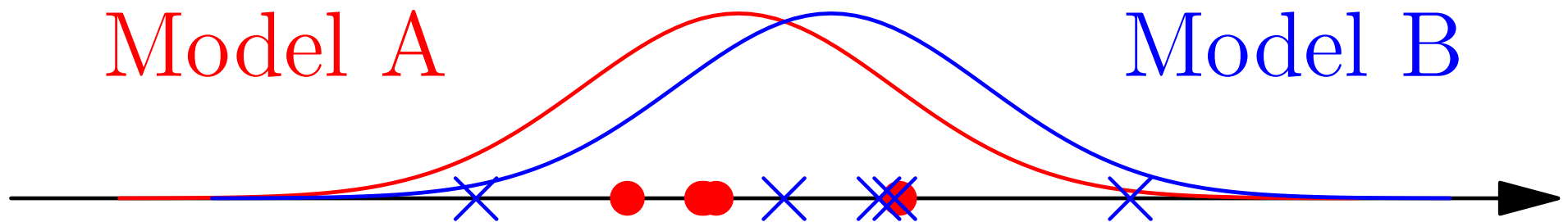
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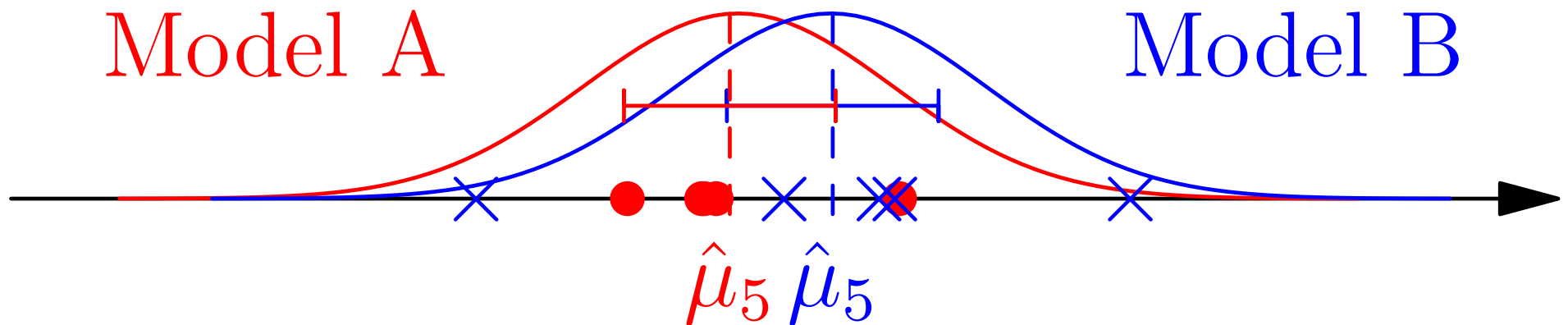
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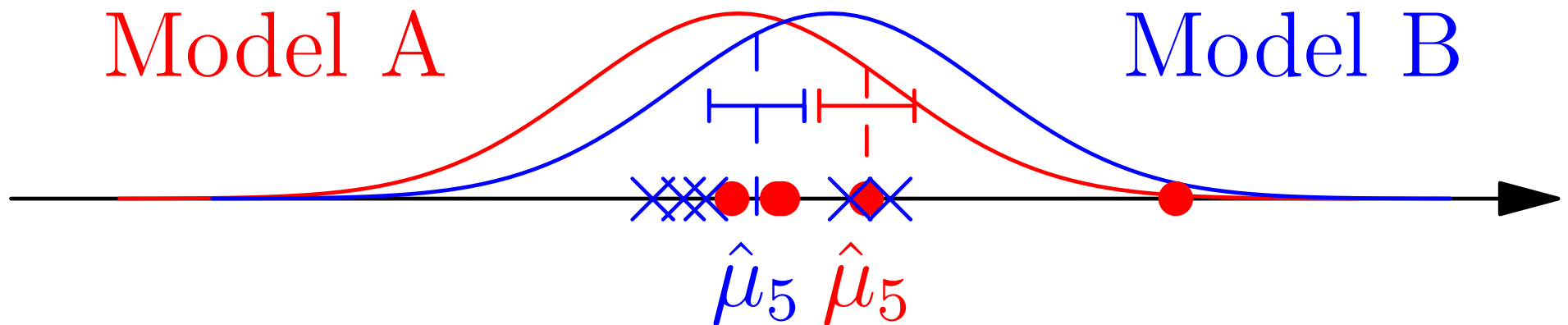
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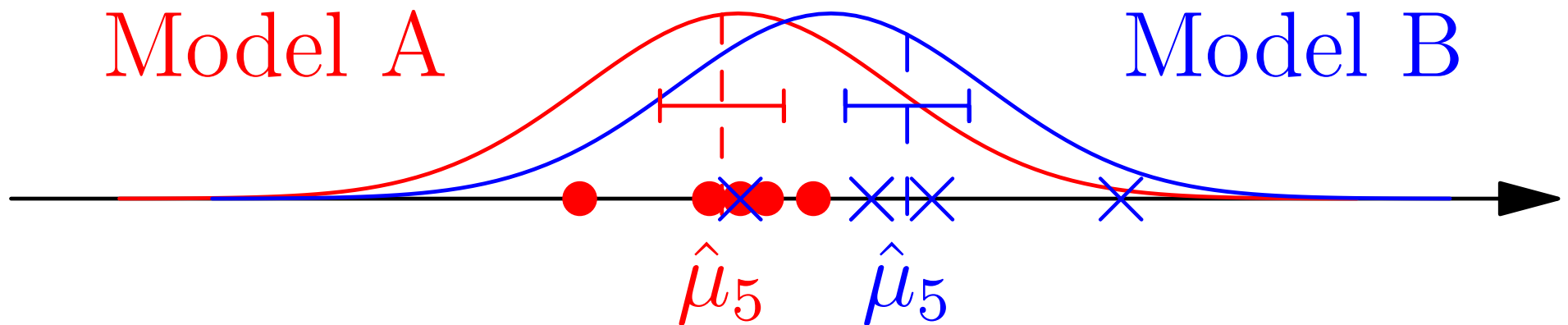
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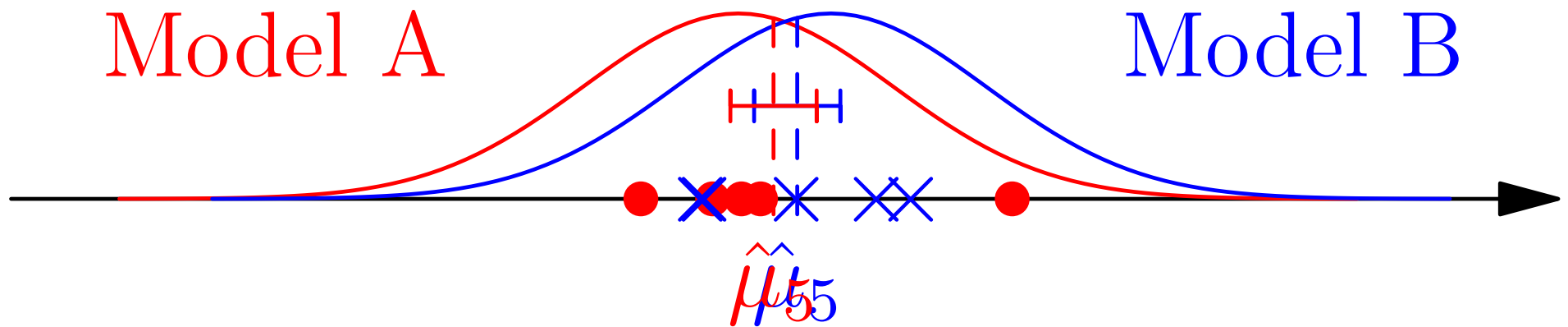
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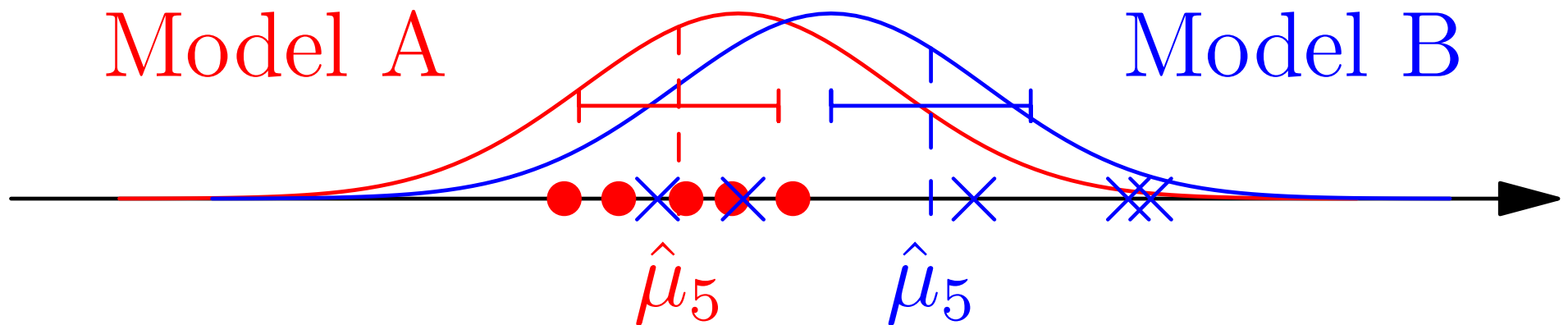
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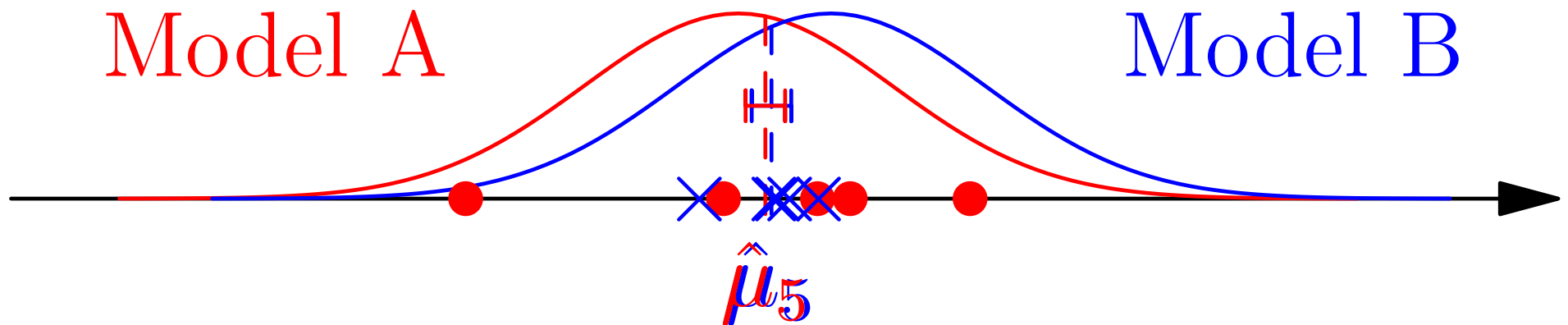
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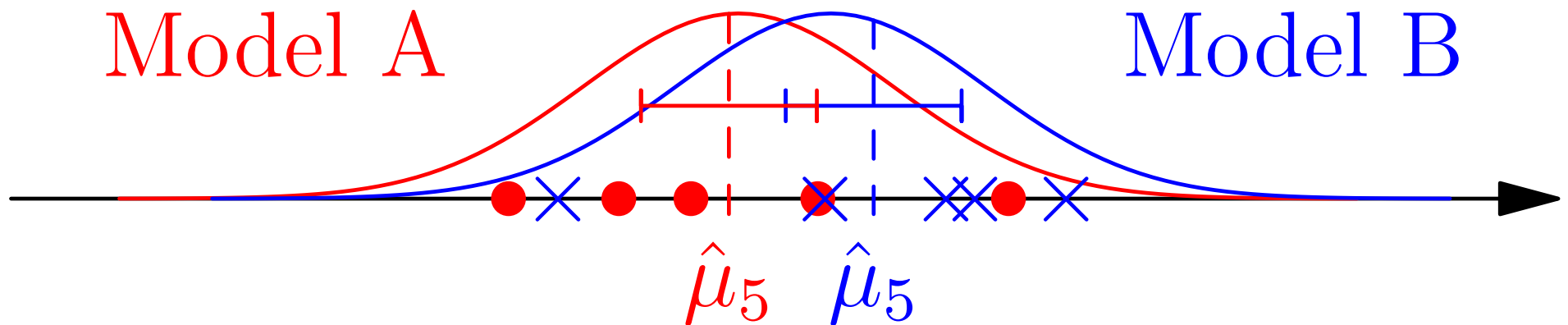
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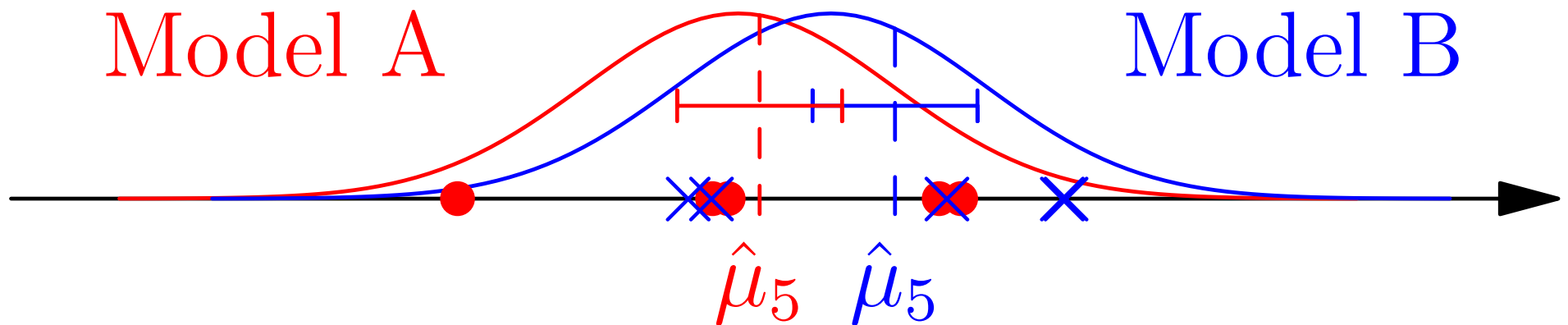
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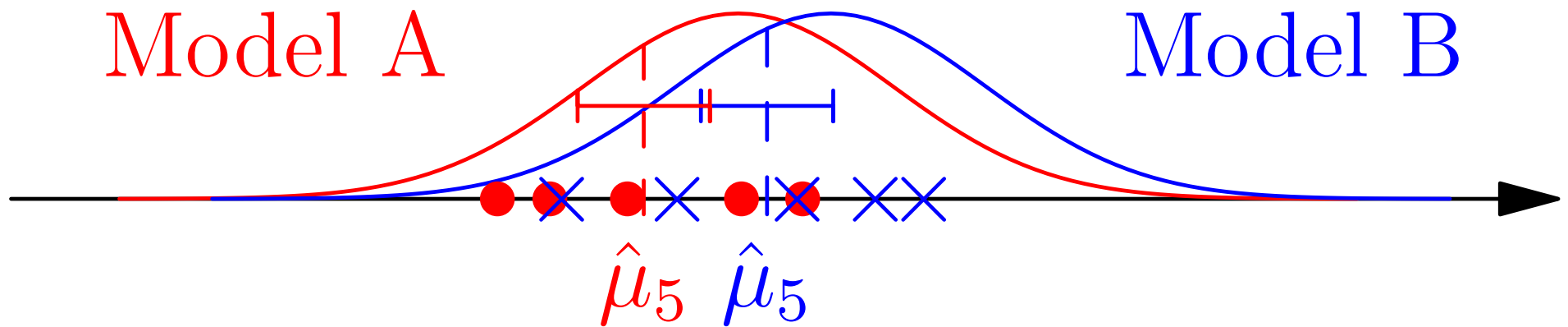
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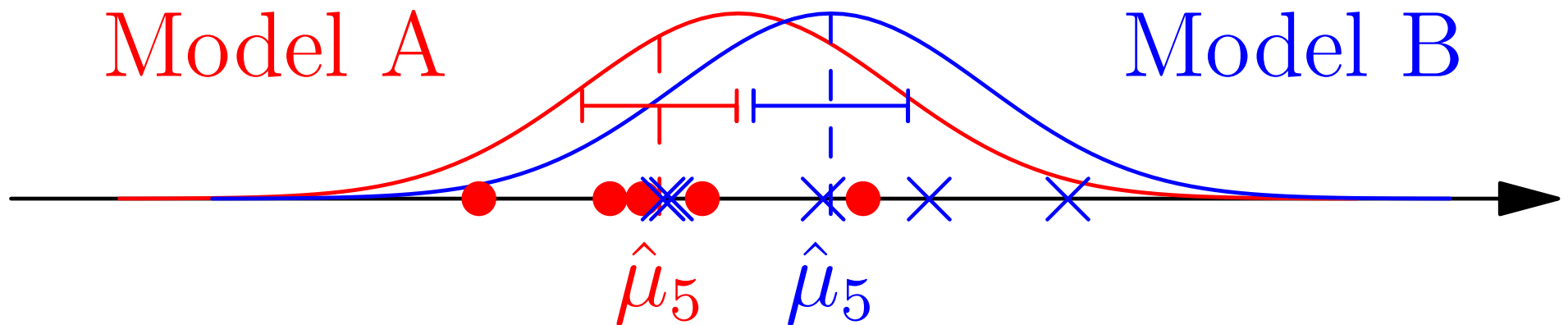
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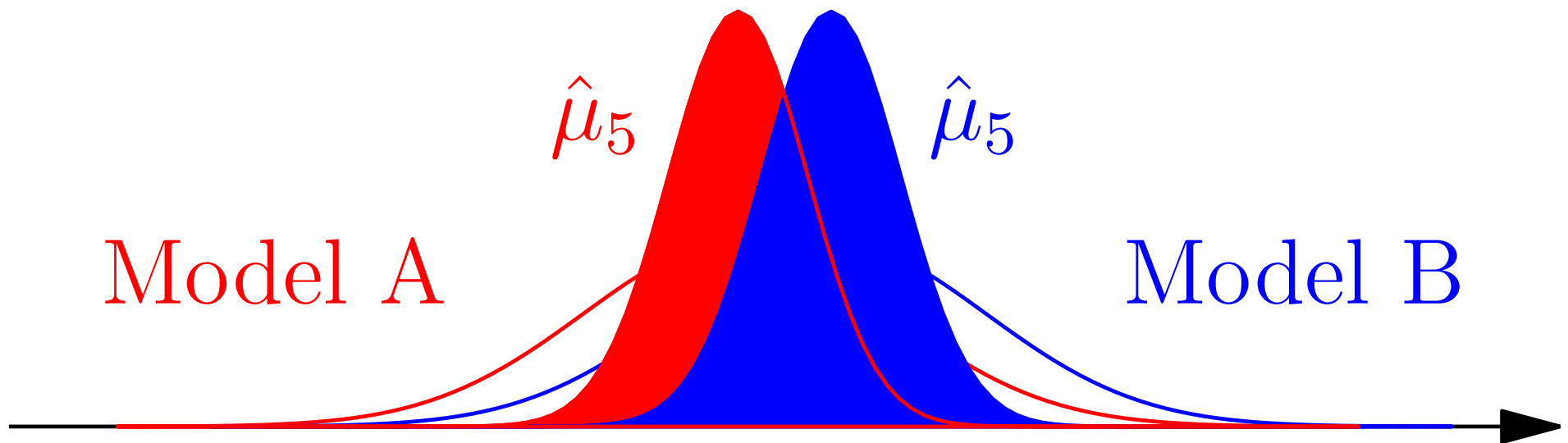
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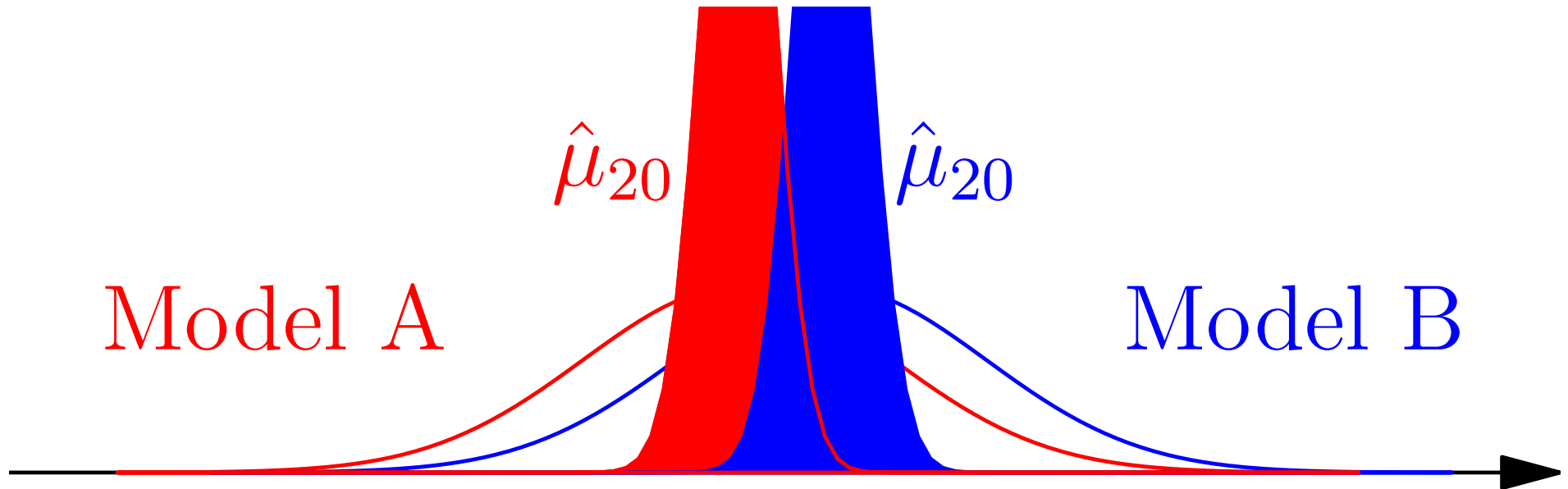
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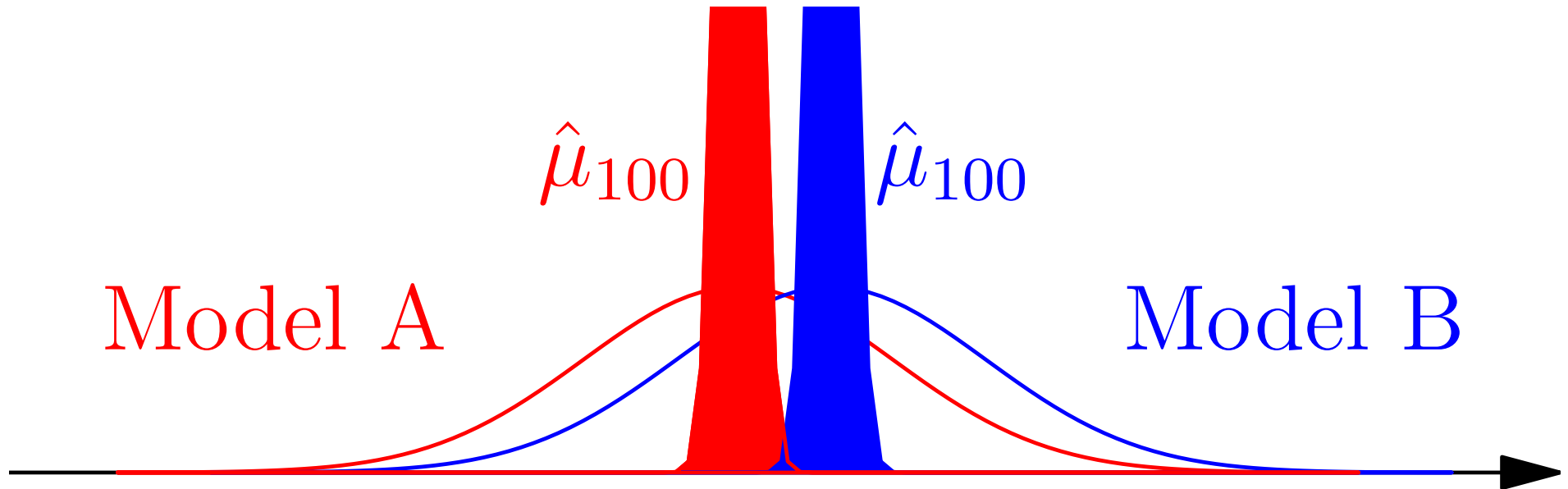
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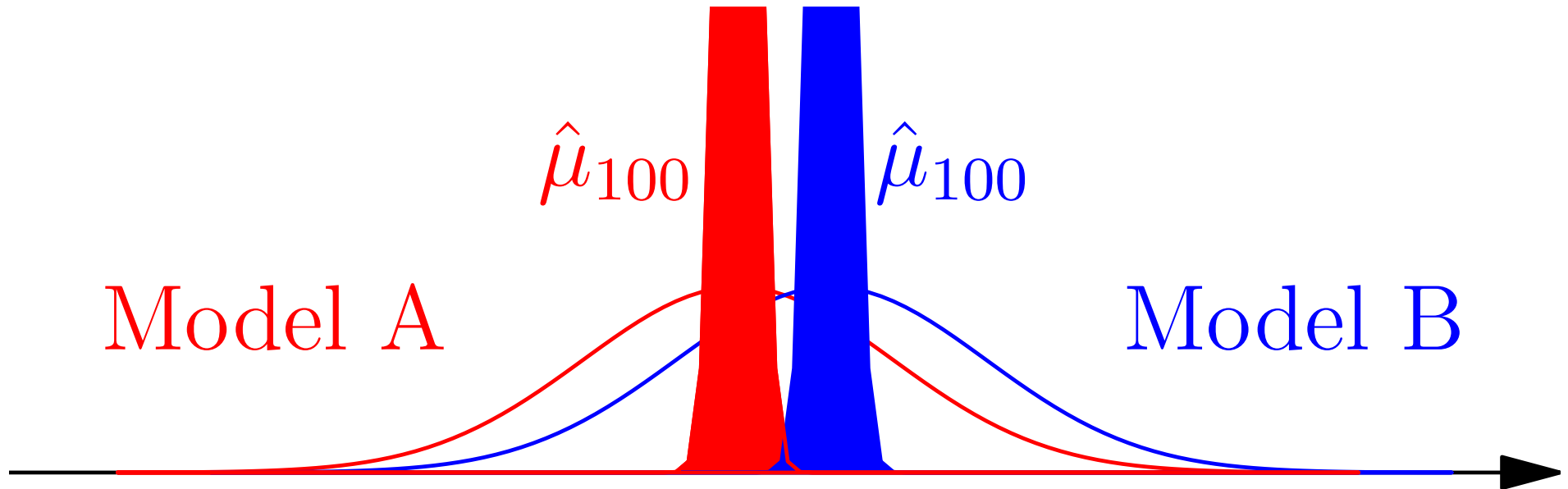
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The more data we have the less likely we will make a mistake

Quantifying Uncertainty

- We want a way of quantifying the chance that model B is better than model A
- To do this we can consider a **null hypothesis**

We suppose that model A and B have the same performance

- We then ask, under the null hypothesis, what is the chance of obtaining a gap in performance at least as big as the one we observed
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- He worked with the null hypothesis that the distributions for A and B are normal with the same means and standard deviations
- The slightly more common test is Welch's t-test where the null hypothesis is that the two datasets are normally distributed with the same means but different standard deviations
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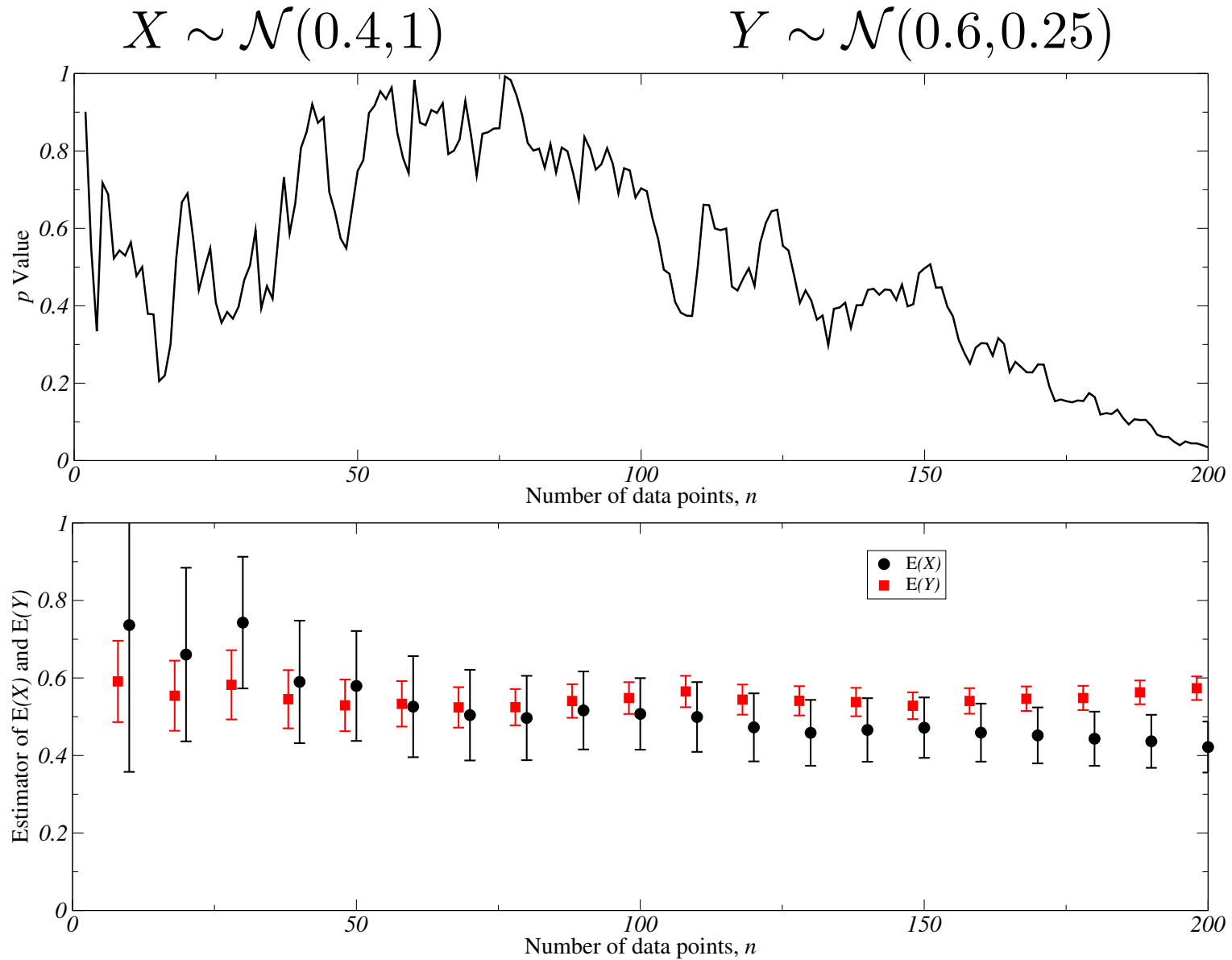
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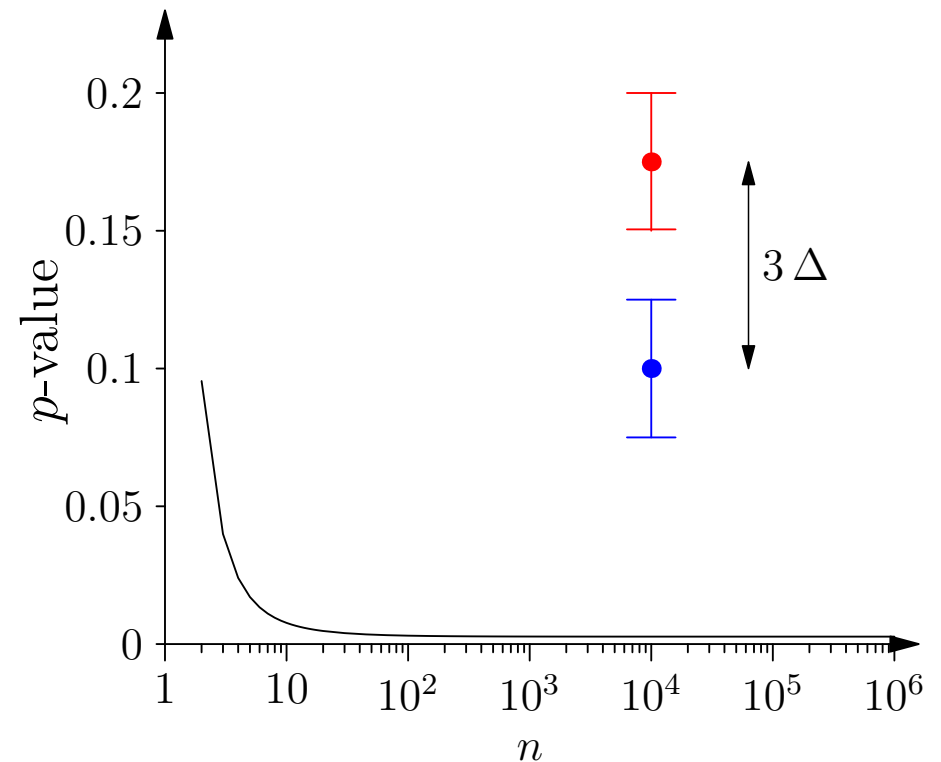
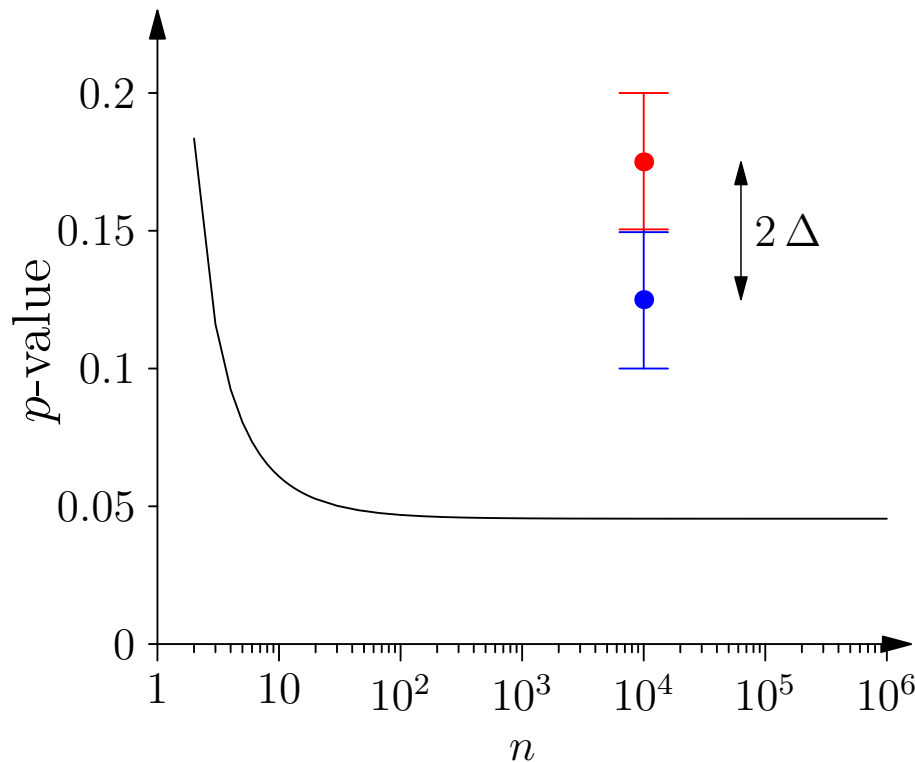
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Example

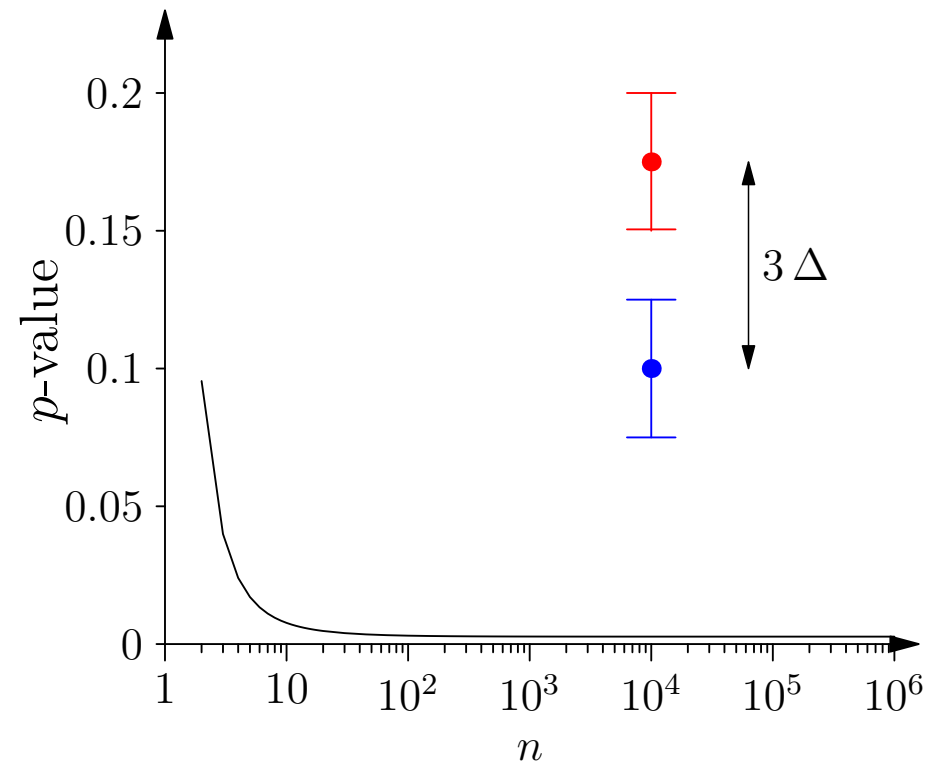
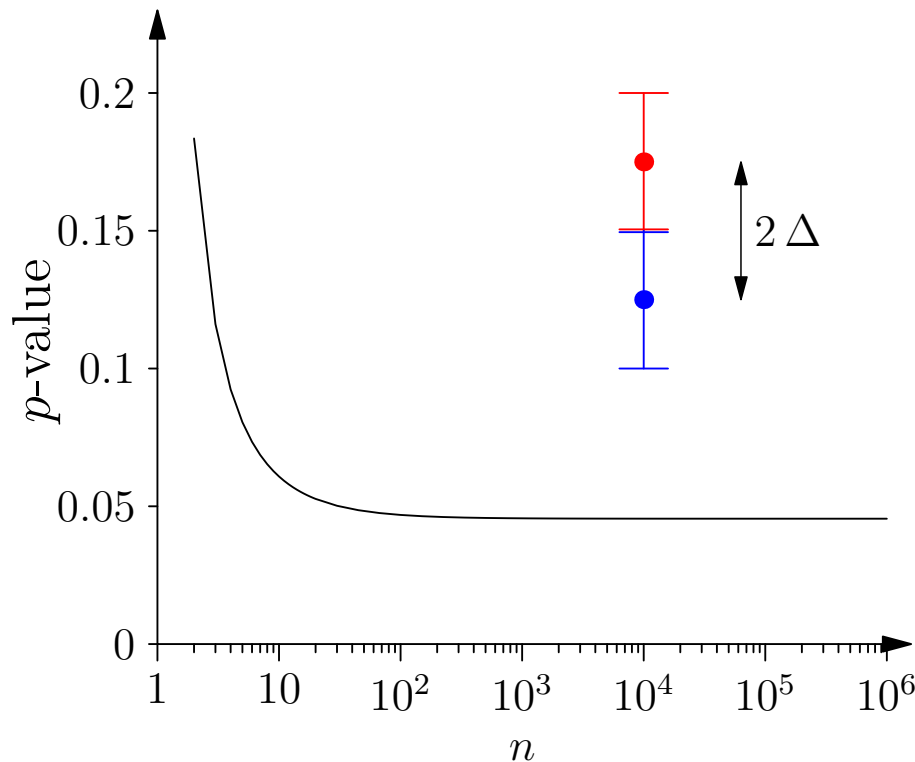


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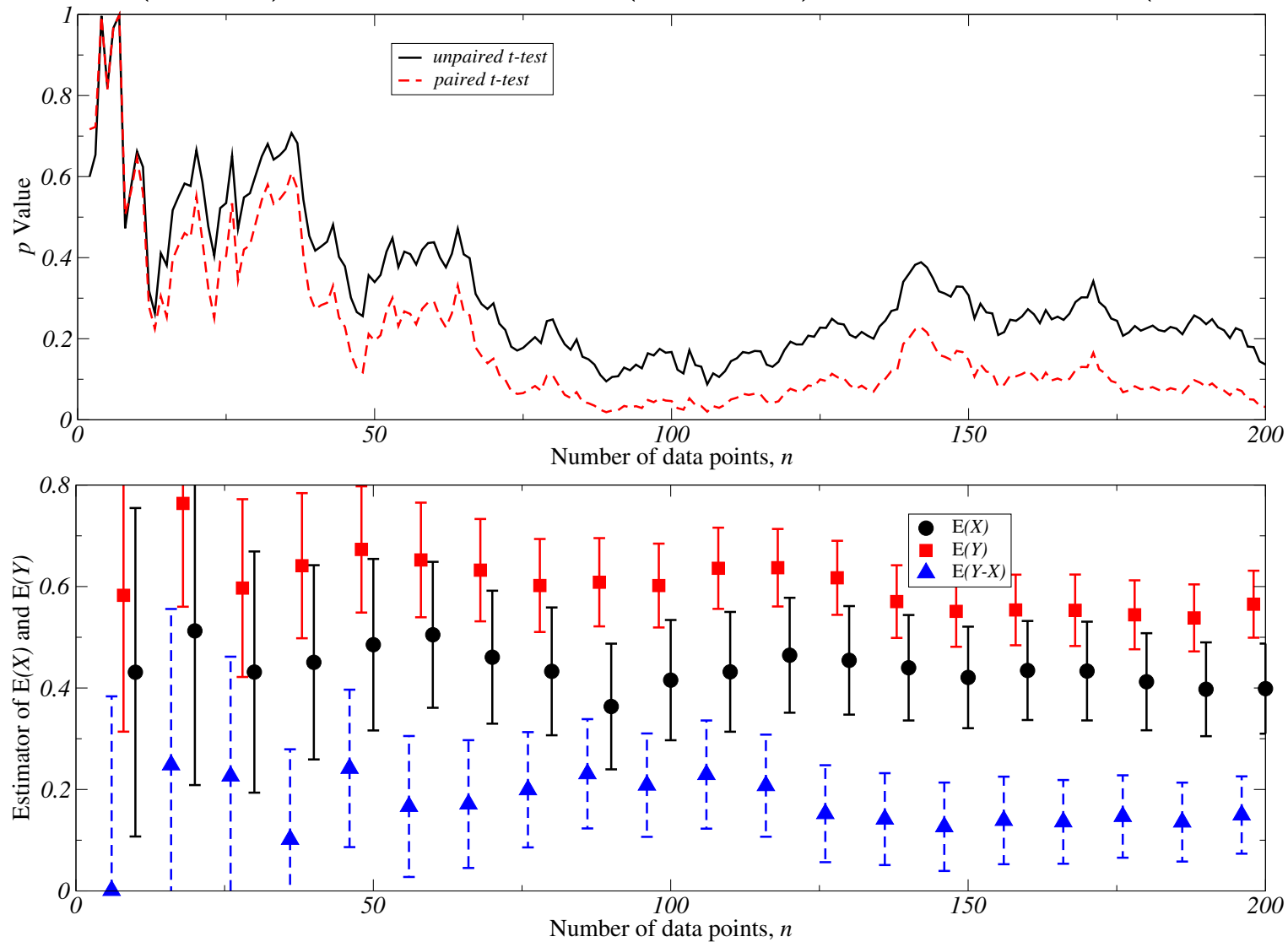
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Paired T-Test Example

$$\begin{aligned} X &\sim Z + X_1 & Y &\sim Z + Y_1 \\ Z &\sim \mathcal{N}(0, 0.8) & X_1 &\sim \mathcal{N}(0.4, 0.2) & Y_1 &\sim \mathcal{N}(0.6, 0.2) \end{aligned}$$



Cheating Significant Tests

- You should only run your t-test once
- If you run multiple t-tests until you get one with $p < 0.05$ then you have cheated
- With a p value of 0.05 you expect to find the observed t value given the null hypothesis once every 20 trials
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- The t -test results follow from a beautiful, but difficult calculation
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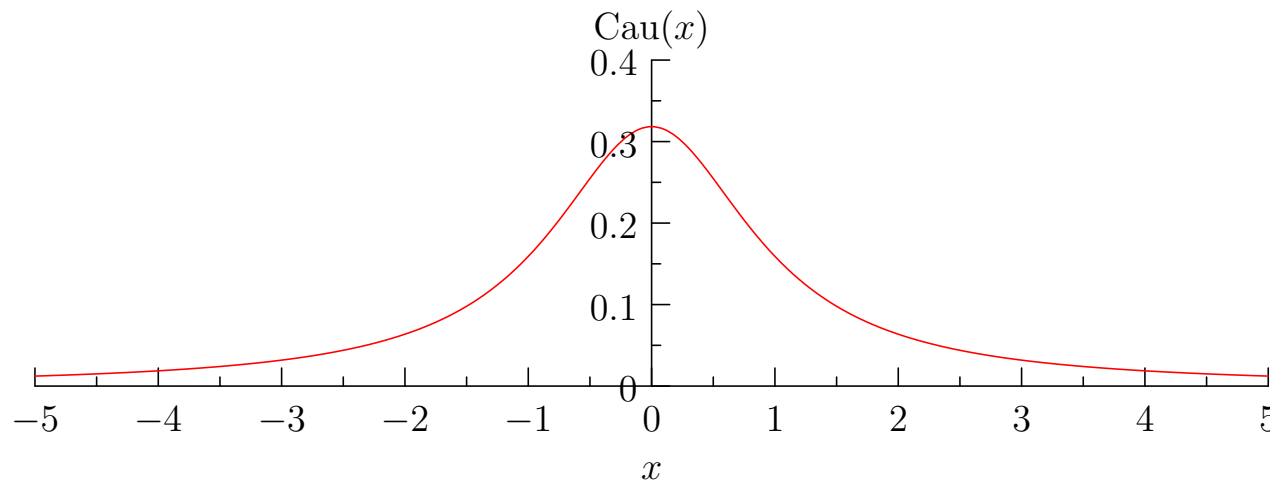
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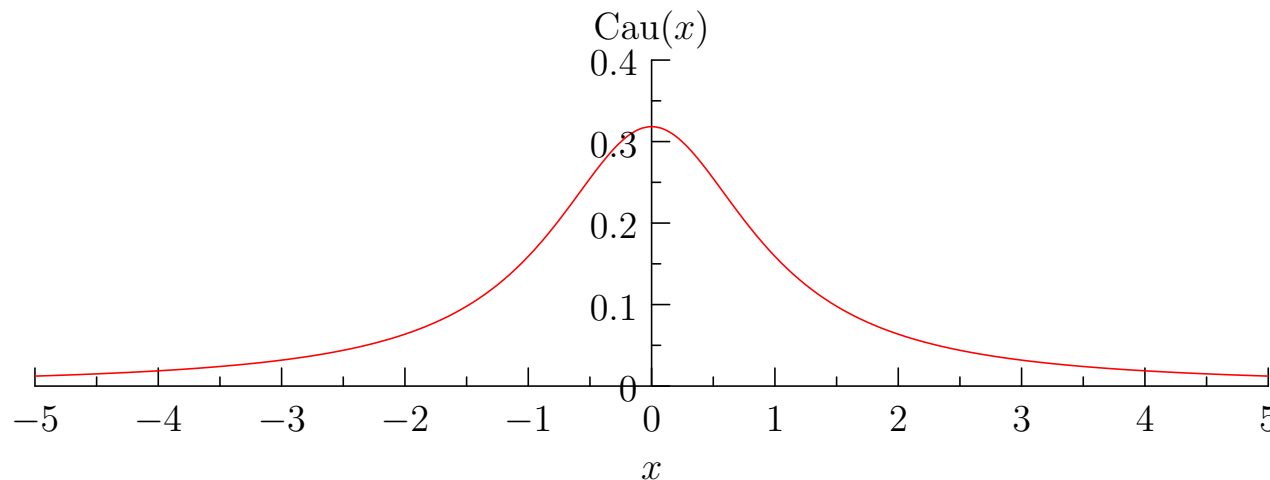
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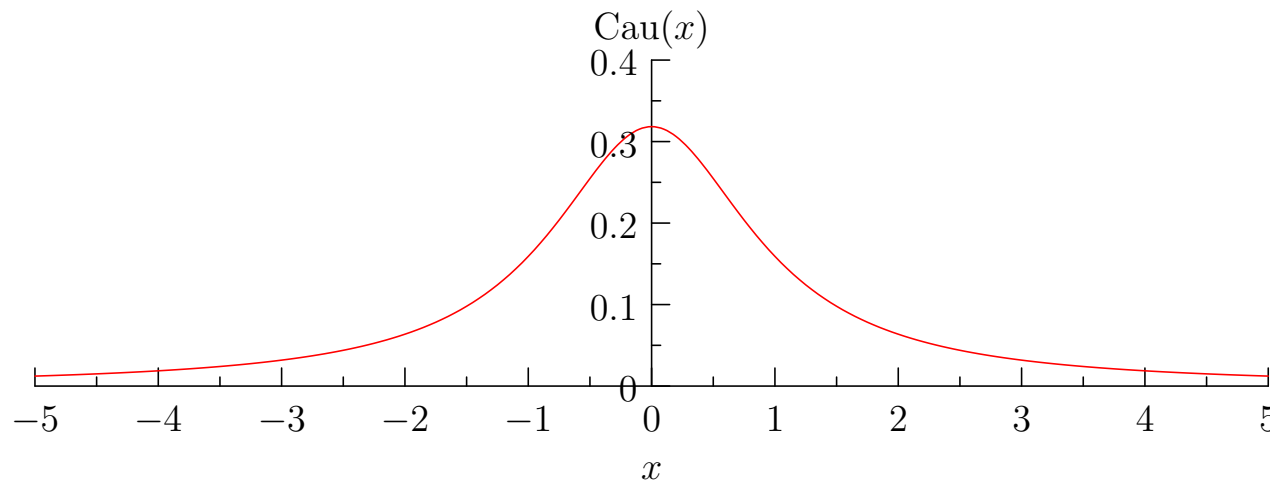
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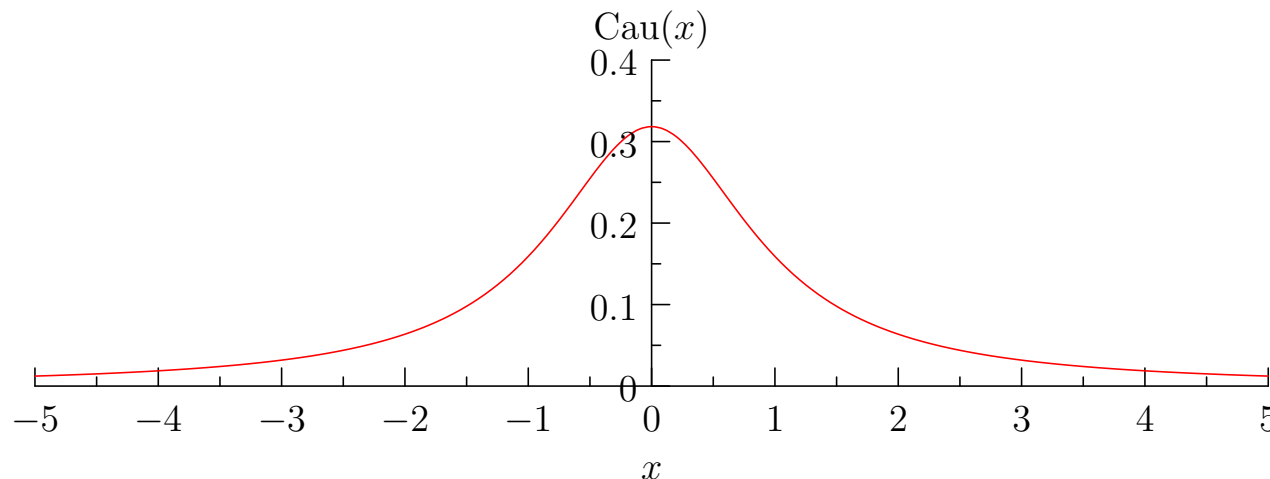
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